Answer **one** question in this section in the spaces provided.

8 **(a)** **(i)** Define simple harmonic motion.

.....

.....

..... [2]

(ii) A dock has a tidal entrance at which the water is 10 m deep at 12 noon, when the tide is at its lowest. The water is 30 m deep at its highest, which occurs next at 6.15 pm. A tanker, needing a depth of 15 m, requires entering the dock as soon as possible that afternoon.

Calculate the earliest time it could just clear the dock entrance. Show your working.

time .....[6]

(iii) State what you have assumed about the motion of the tide.

..... [1]

(b) Fig. 8.1 shows a horizontal copper wire placed between the opposite poles of a permanent magnet. The wire is in a state of tension and is clamped at each end.

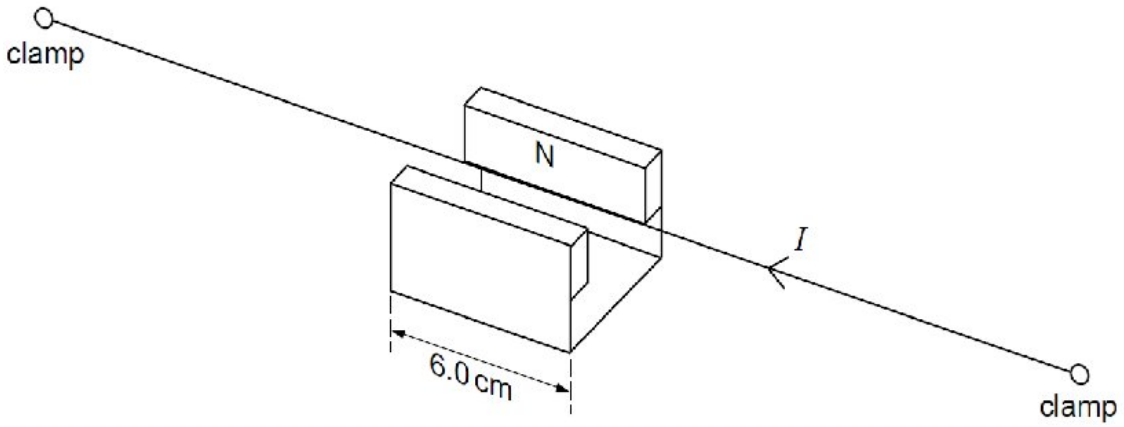


Fig. 8.1

(i) An alternating current I of constant amplitude and variable frequency is passed through the wire causing the wire to oscillate. Fig. 8.2 shows how the acceleration of the wire at the centre point between the poles varies with time when the frequency of the current is at the fundamental natural frequency of the wire.

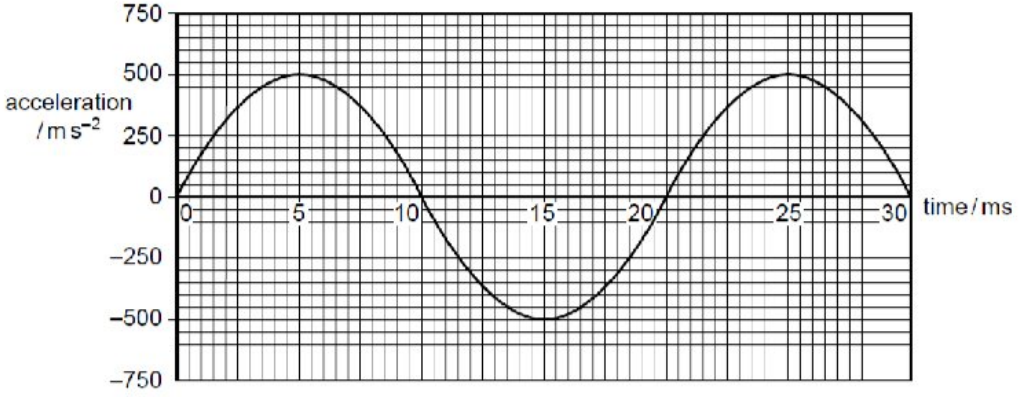


Fig. 8.2

1. Calculate the frequency of the alternating current.

frequency = ... Hz [2]

2. Explain whether the maximum acceleration at all points on the wire between the poles is the same or not. A sketch may help your answer.

.....

.....

.....

..... [3]

(ii) The amplitude of vibration of the wire at the centre point between the poles varies with the frequency of the current as shown in Fig 8.3.

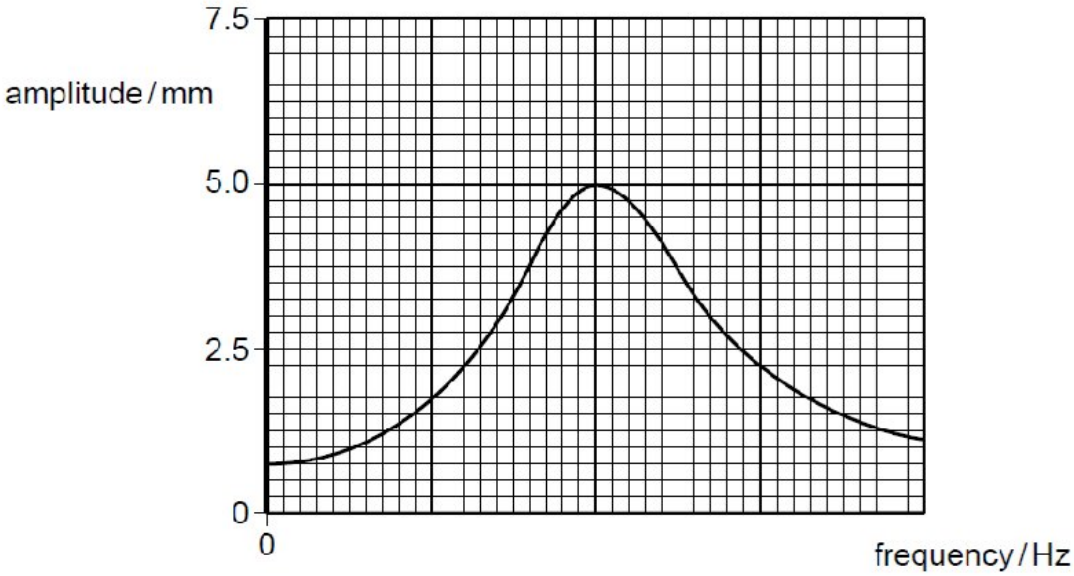


Fig. 8.3

1. State the term used to describe this effect.

..... [1]

2. Add the scale to the frequency axis on Fig 8.3. [1]

3. A student sticks some small pieces of paper to the wire to view the vibration of the wire more easily. This increases the damping of the vibration.

Explain what is meant by damping and on Fig. 8.3 above, sketch the graph you would expect to see for greater damping.

.....

..... [4]

[Total: 20]